

## Finding the base of a triangle using the area and height

The formula for the area of a triangle can also be used to find the length of the base, if the area and height are known. Given the area and height of the triangle, the formula needs to be rearranged to find the length of the triangle's base.

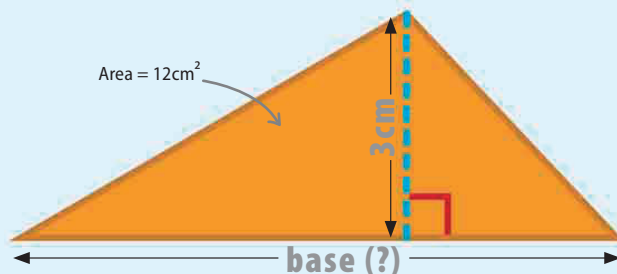
**First, write down the formula** for the area of a triangle. The formula states that the area of a triangle is equal to  $\frac{1}{2}$  multiplied by the length of the base, multiplied by the height.

**Substitute the known values** into the formula. Here the values of the area ( $12\text{cm}^2$ ) and the height ( $3\text{cm}$ ) are known.

**Simplify the formula** as far as possible, by multiplying the  $\frac{1}{2}$  by the height. This answer is 1.5.

**Make the base the subject** of the formula by rearranging it. In this example both sides are divided by 1.5.

**Work out the final answer** by dividing 12 (area) by 1.5. In this example, the answer is 8cm.



$$\text{area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$12 = \frac{1}{2} \times \text{base} \times 3$$

$$12 = 1.5 \times \text{base}$$

$\frac{1}{2} \times 3 = 1.5$   
base is unknown

$$\frac{12}{1.5} = \text{base}$$

as base was multiplied by 1.5, divide this side by 1.5 to cancel out the 1.5s and leave base on its own on this side  
as the other side has been divided by 1.5, this side must also be divided by 1.5

$$\text{base} = 8\text{cm}$$

## Finding the vertical height of a triangle using the area and base

The formula for area of a triangle can also be used to find its height, if the area and base are known. Given the area and the length of the base of the triangle, the formula needs to be rearranged to find the height of the triangle.

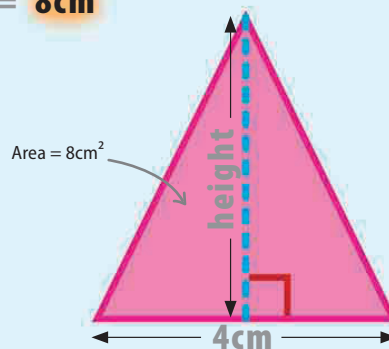
**First, write down the formula.** This shows that the area of a triangle equals  $\frac{1}{2}$  multiplied by its base, multiplied by its height.

**Substitute the known values** into the formula. Here the values of the area ( $8\text{cm}^2$ ) and the base ( $4\text{cm}$ ) are known.

**Simplify the equation** as far as possible, by multiplying the  $\frac{1}{2}$  by the base. In this example, the answer is 2.

**Make the height the subject** of the formula by rearranging it. In this example both sides are divided by 2.

**Work out the final answer** by dividing 8 (the area) by 2 ( $\frac{1}{2}$  the base). In this example the answer is 4cm.



$$\text{area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$8 = \frac{1}{2} \times 4 \times \text{height}$$

height is unknown

$$8 = 2 \times \text{height}$$

$\frac{1}{2} \times 4 = 2$

$$\frac{8}{2} = \text{height}$$

this side must be divided by 2 to cancel out the 2s and leave height on its own on this side  
as the other side has been divided by 2, this side must also be divided by 2

$$\text{height} = 4\text{cm}$$